

Homework 2: Rates, ratios, and discounting

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Present value

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present_value([10, 15, 20], 0.10) = 36.51  
bond_price(1000, 0.04, 10, 0.04) = 1000.00
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10-year bond, 4% coupon, face 1000
market rate 2.00%: price 1179.65
market rate 4.00%: price 1000.00
market rate 4.96%: price 925.73



The price curve bends (it is convex): the price rises about \$180 when the market rate falls from 4% to 2% but only falls about \$74 when it rises to 4.96%, so a bond gains more when rates fall than it loses when they rise by the same amount.

Beta from real data

Tickers: AAPL (technology), XOM (energy), JNJ (health care), benchmark SPY. Five years of daily adjusted closes.

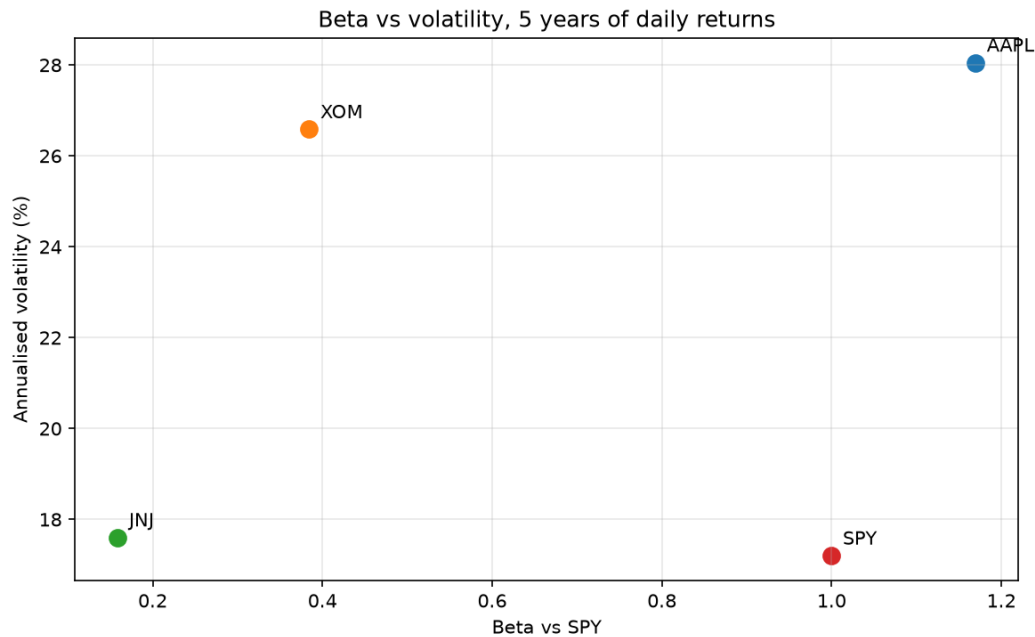
Trading days:

AAPL: 1255
XOM: 1255
JNJ: 1255
SPY: 1255

Ticker	Return	Volatility	Beta
AAPL	21.56%	28.05%	1.17
XOM	27.09%	26.58%	0.38
JNJ	14.57%	17.60%	0.16

SPY	13.98%	17.20%	1.00
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Return and volatility are annualised from daily returns (mean x 252, std x sqrt(252)). Beta = $\text{cov}(\text{stock}, \text{SPY}) / \text{var}(\text{SPY})$, computed on daily returns, not prices. SPY against itself gives 1.00.



Written

Yes, the two rankings agree in this sample: AAPL, then XOM, then JNJ, on both volatility and beta. But the agreement hides a big gap in size. XOM is where beta and volatility tell different stories: its volatility is 26.6%, nearly as high as AAPL's 28.1% and far above SPY's 17.2%, yet its beta is only 0.38, against 1.17 for AAPL. Volatility measures total variation, and beta measures only the part that moves with the market. XOM's swings come mostly from things specific to it or its sector, such as oil prices, so most of its risk is idiosyncratic and can be diversified away in a portfolio. A stock can be jumpy without being sensitive to the market, and beta is the number that captures the market-sensitive part.